RSAConference2015

Singapore | 22-24 July | Marina Bay Sands

CHANGE

Challenge today's security thinking

SESSION ID: SPO-R08

Understanding the Data Breaches of 2014: Did it have to be this way?

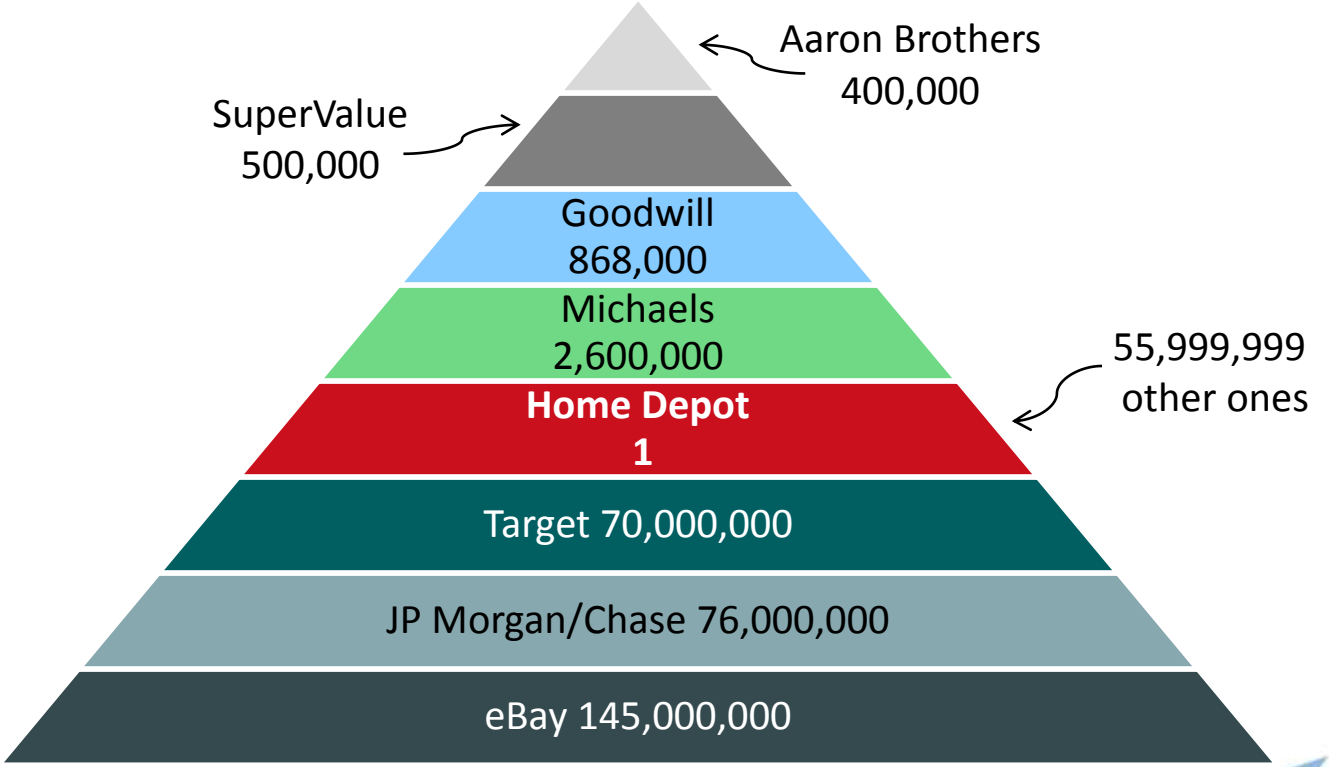
Patrick Grillo

Senior Director, Product Strategy
Fortinet

#RSAC

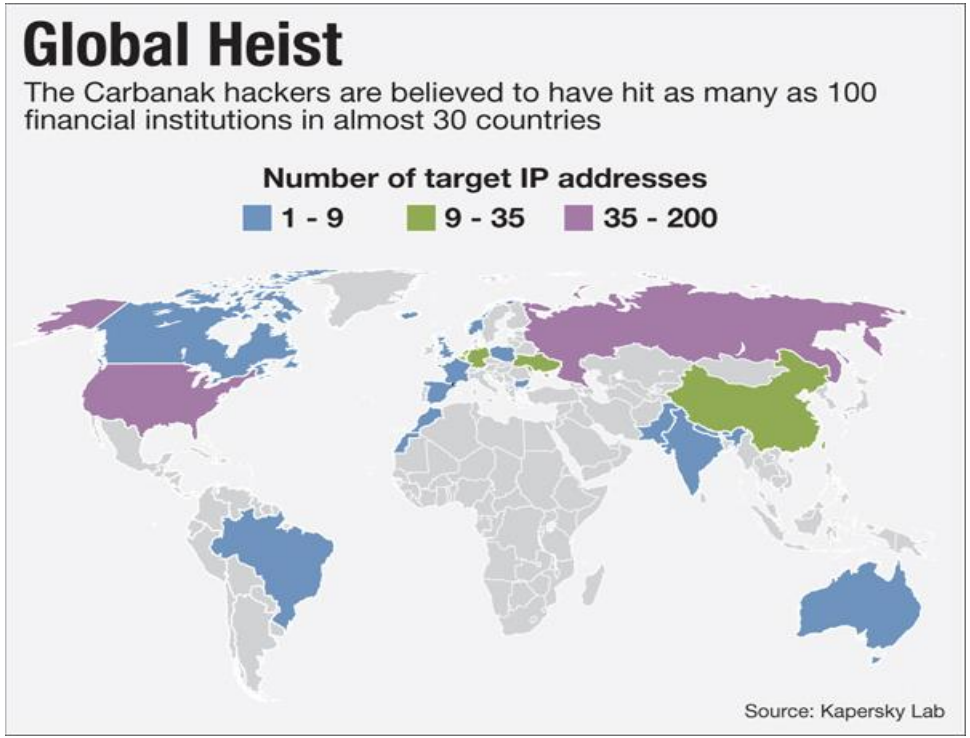
[redacted]

Anyone Want To Add Anything?

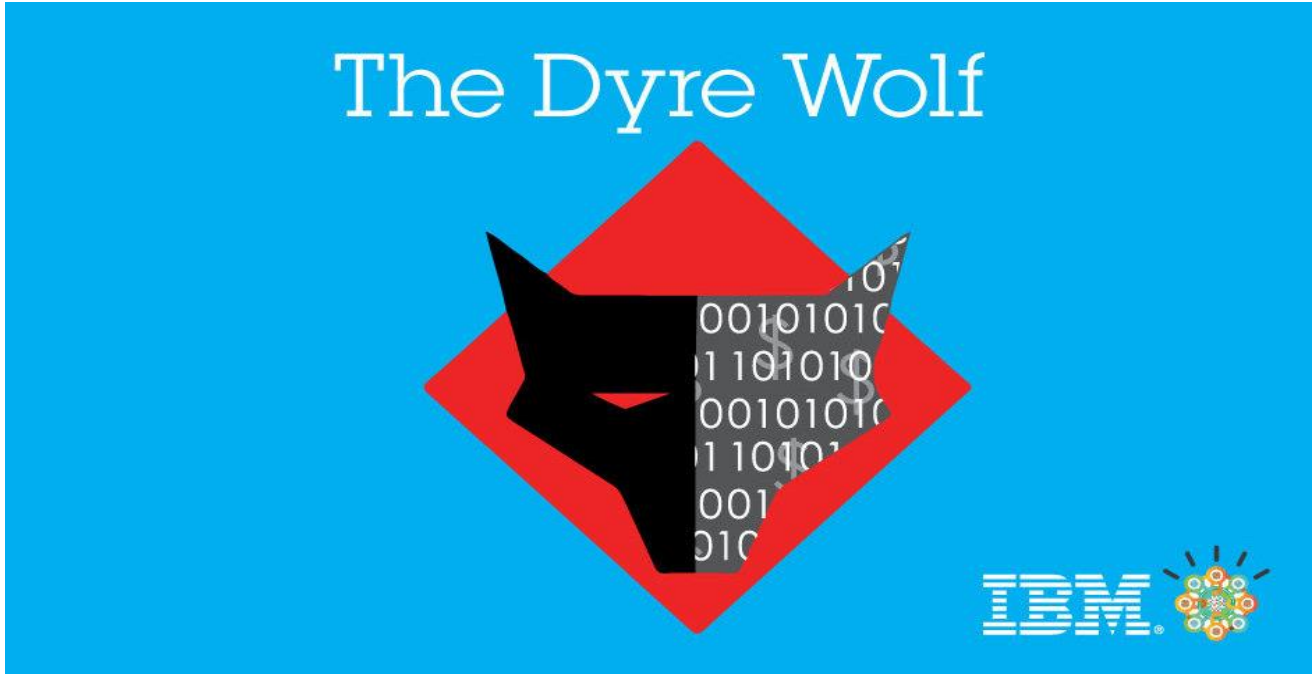


*Milton Security

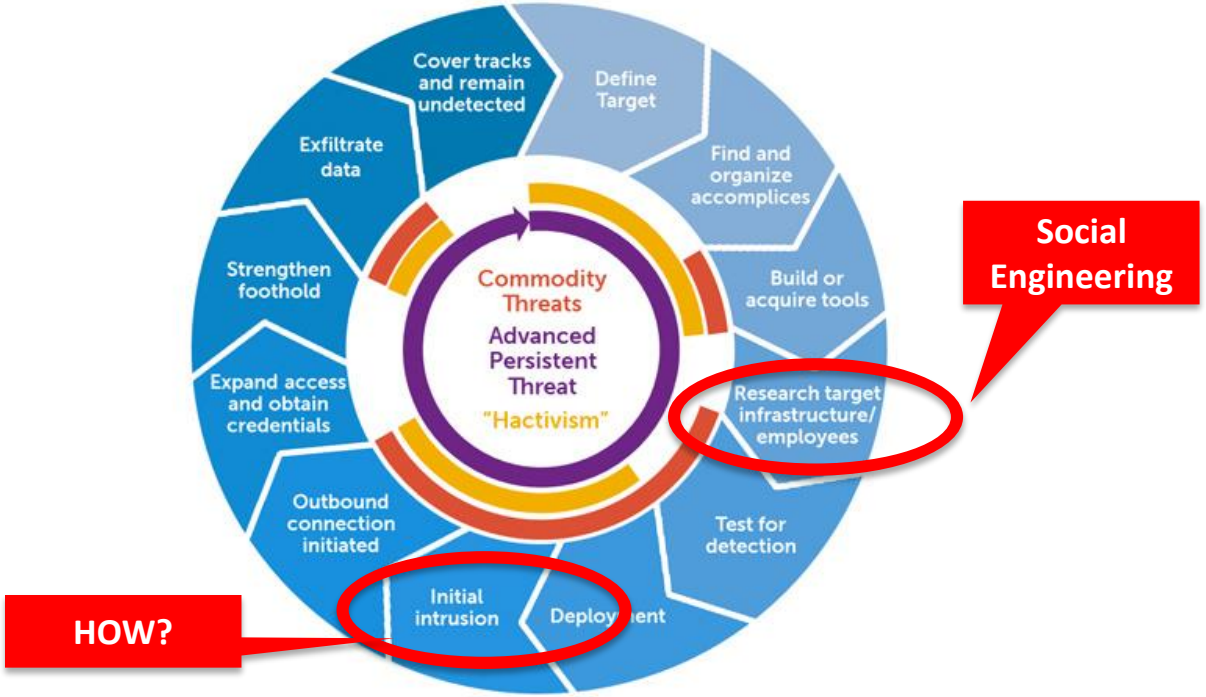
21st Century Bank Robbery



Not Just in the Game of Thrones

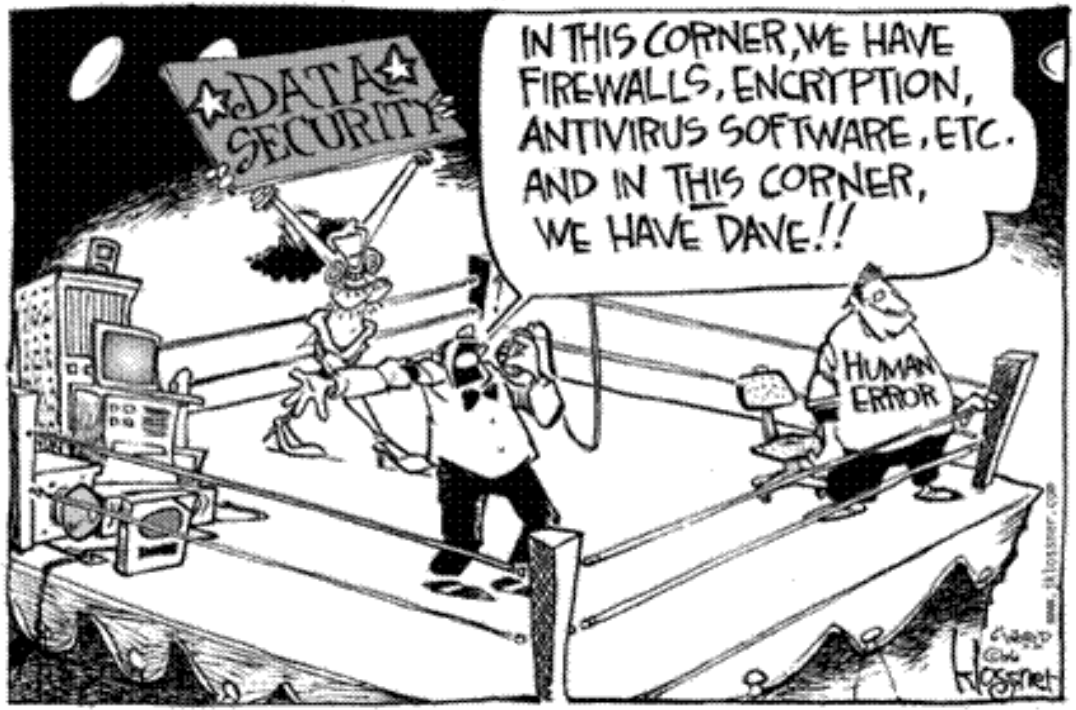


Advanced Persistent Threat Structure



*Wikipedia Commons

Exploiting the Weak Link



The Common Thread

Target -
Outside contractor clicks on phishing email

http://www.bloomberg.com/bw/articles/2014-03-13/target-missed-alarms-in-epic-hack-of-credit-card-data#p2

http://krebsonsecurity.com/2014/02/target-hackers-broke-in-via-hvac-company/

Home Depot
Stolen login credentials from Third party vendor

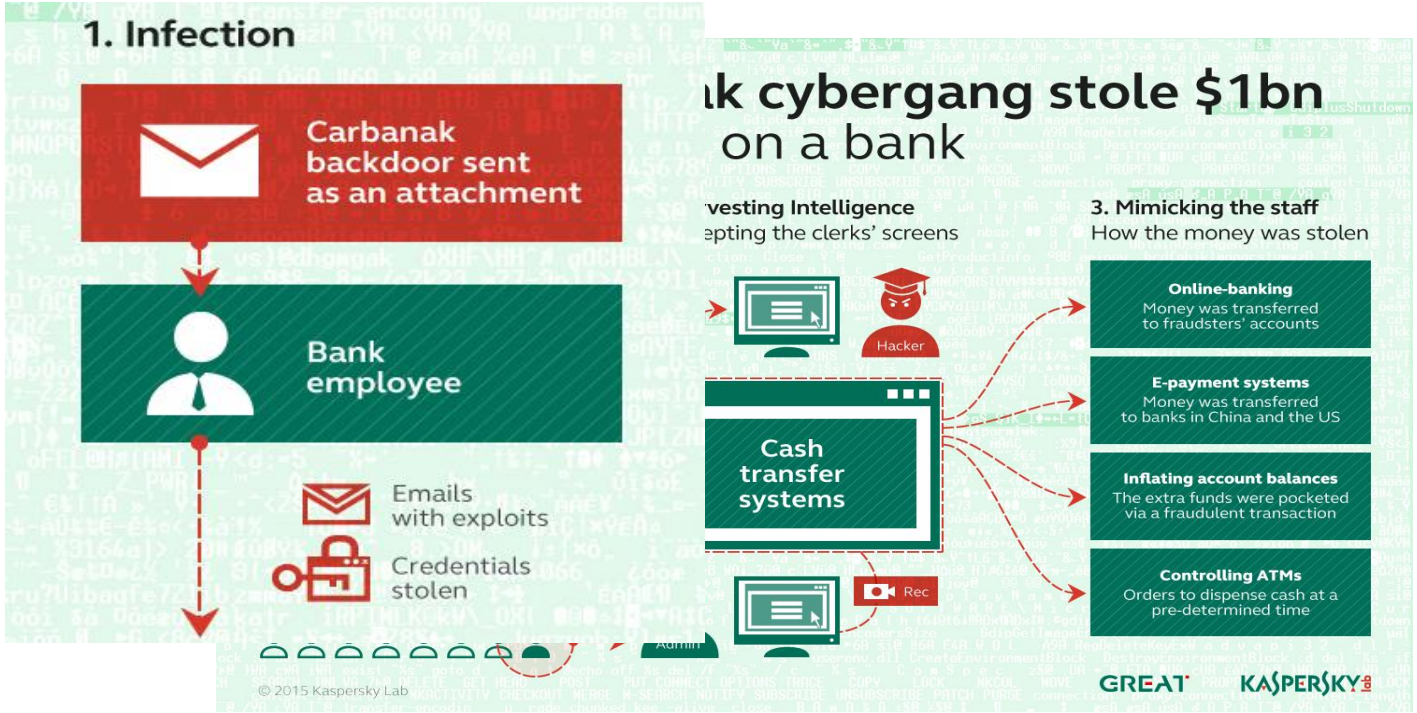
http://krebsonsecurity.com/2014/11/home-depot-hackers-stole-53m-email-addreses/

http://www.eweek.com/security/home-depot-breach-expands-privilege-escalation-flaw-to-blame.html

JP Morgan/Chase –
Stolen employee credentials

http://www.computerworld.com/article/2862578/twofactor-authentication-oversight-led-to-jpmorgan-breach-investigators-reportedly-found.html

Cyber Crime on a Global Basis

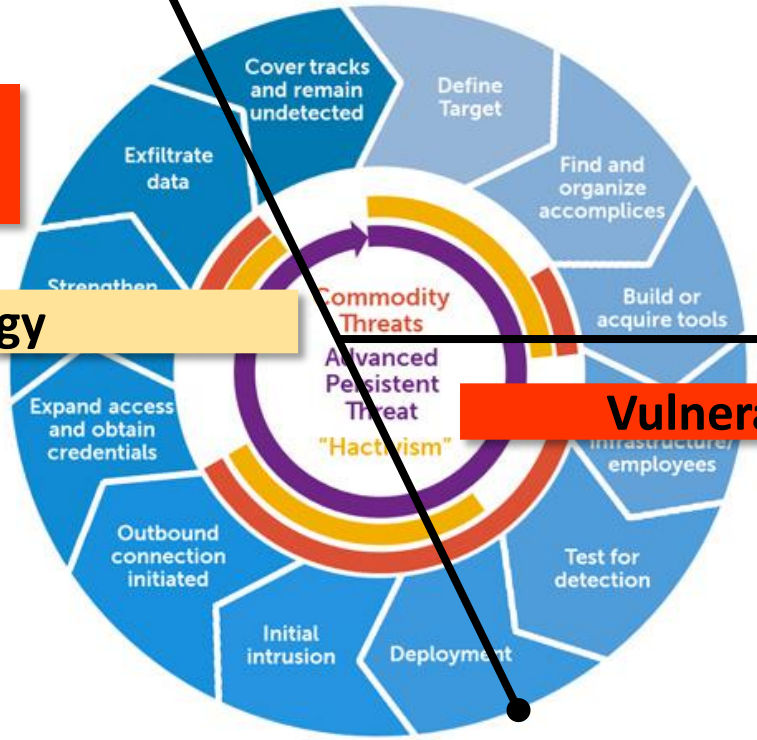


Not Just for Consumers

Let's Take a Closer Look

Best Practices
Network Architecture

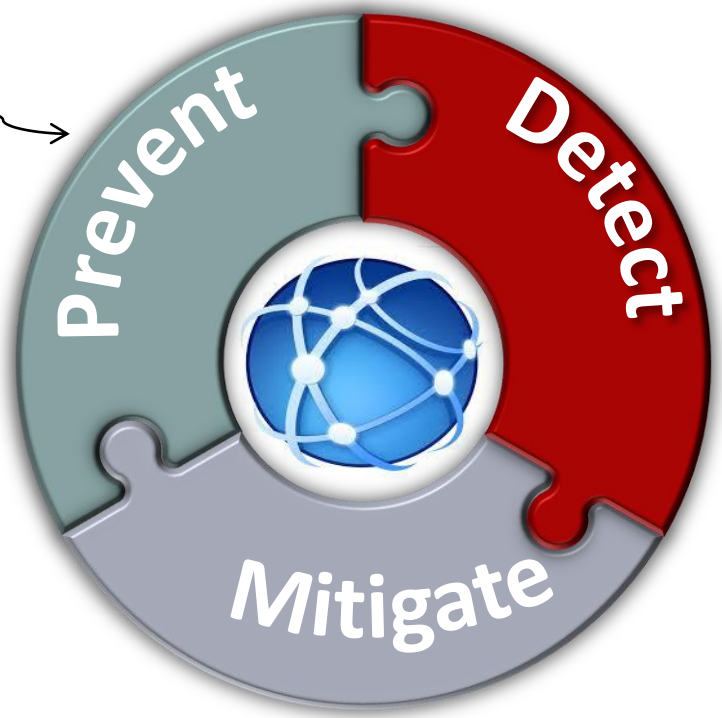
Technology



Vulnerability

The Technology Side of Things

Two Factor Authentication
Anti-Virus
Intrusion Prevention
Secure Email Gateway
Web Application Firewall
End Point Protection



Mixed Messages Only Aid the Hacker

TRENDING: iPhone 7 rumor rollup · Windows 10 vs. Windows 8 · BYOD debate is not over · Resources/White Papers

NETWORKWORLD Most read:

Home > Security

INSIDER

Antivirus doesn't work. So why are you still using it?

By Bob Violino | Follow

Network World | Apr 6, 2015 2:53 AM PT

For years, companies have relied on antivirus (AV) software to help detect, prevent and remove malicious code before it becomes a problem.

But standalone AV is no longer effective at stopping today's increasingly sophisticated barrage of key loggers, backdoors, rootkits, Trojan horses, worms and spyware.

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Other Insider Recommendations

The Technology Side of Things

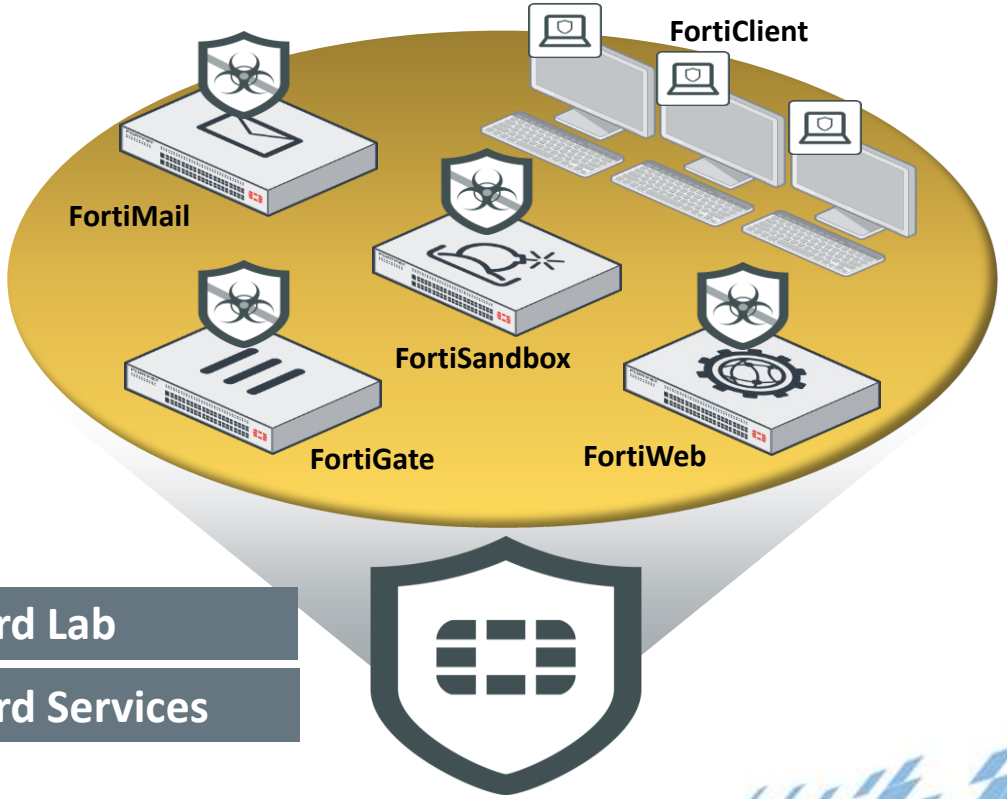
Two Factor Authentication
Anti-Virus
Intrusion Prevention
Secure Email Gateway
Web Application Firewall
End Point Protection

Botnet Detection
IP & Client Reputation
Sandboxing

People
Process
Technology

Threat Intelligence
Zero Day Research
Continuous updates

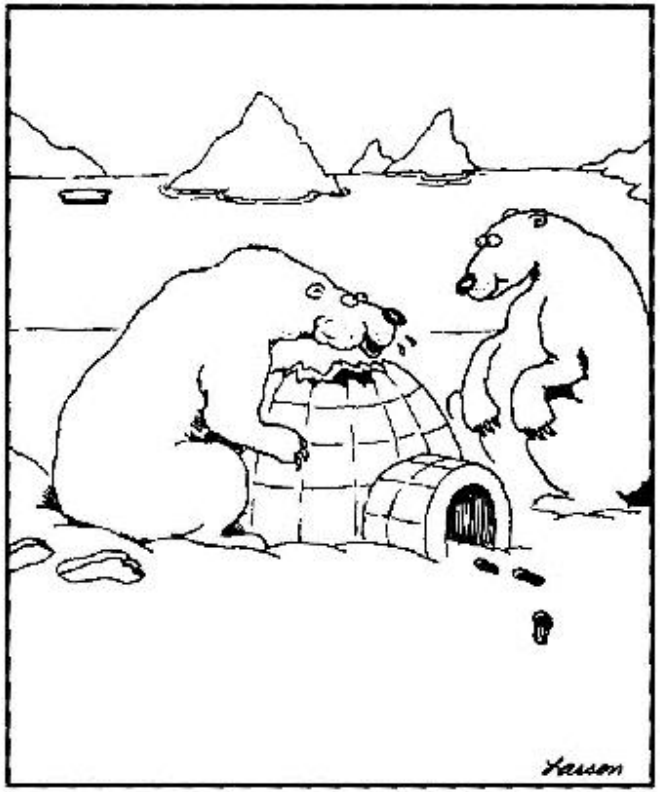
Advanced Threat Protection



FortiGuard Lab

FortiGuard Services

Protecting Deeper Into the Network

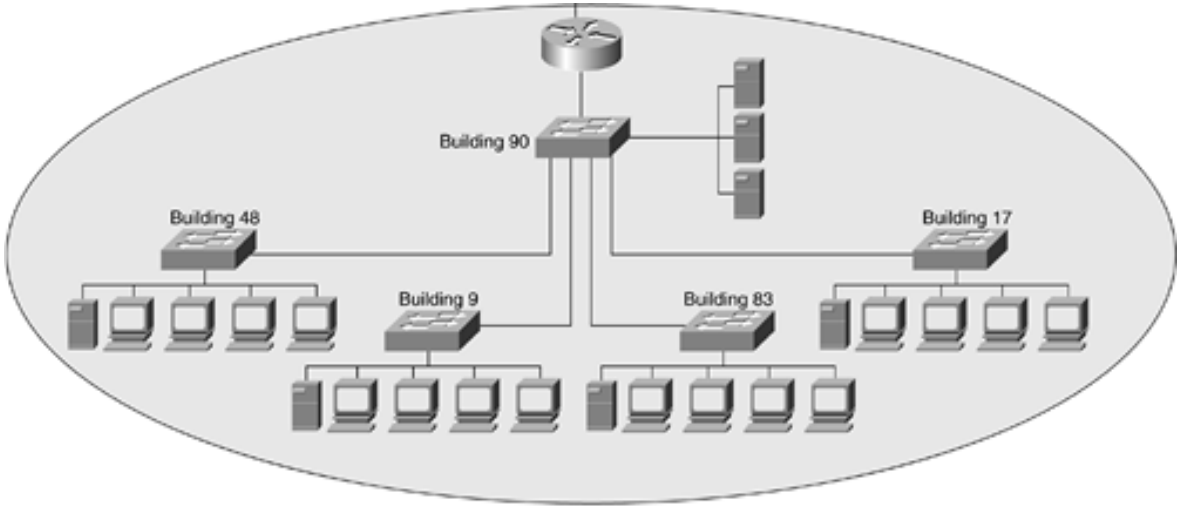


"Oh, hey! I just love these things! ... Crunchy on the outside and a chewy center!"

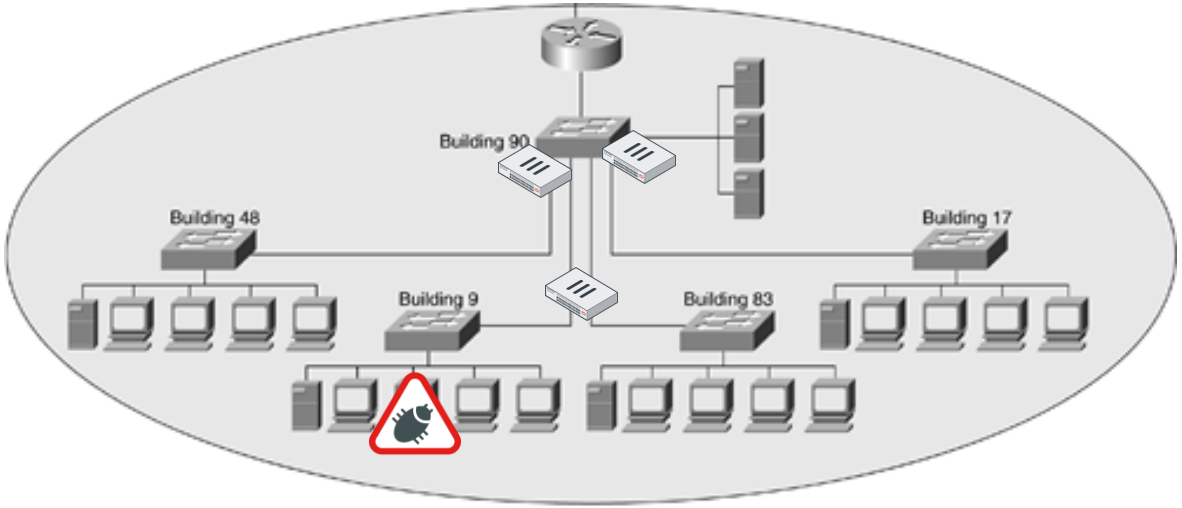
#TEISS15: How to control shadow IT initiatives

What do Igloos and the networks have in common? Patrick Grillo, senior director of solutions marketing at Fortinet, says "they…

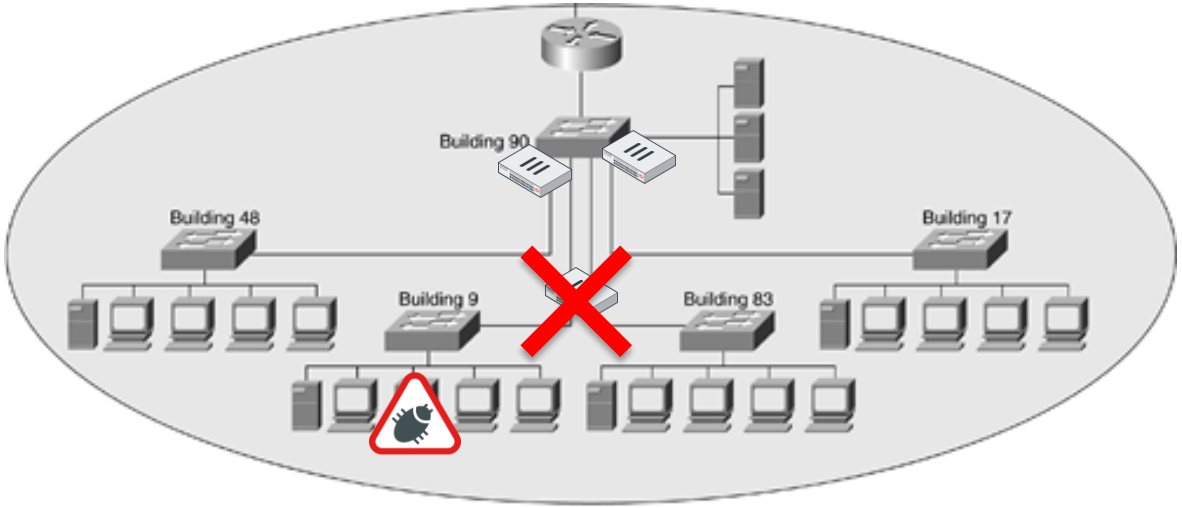
Internal Segmentation For Greater Control



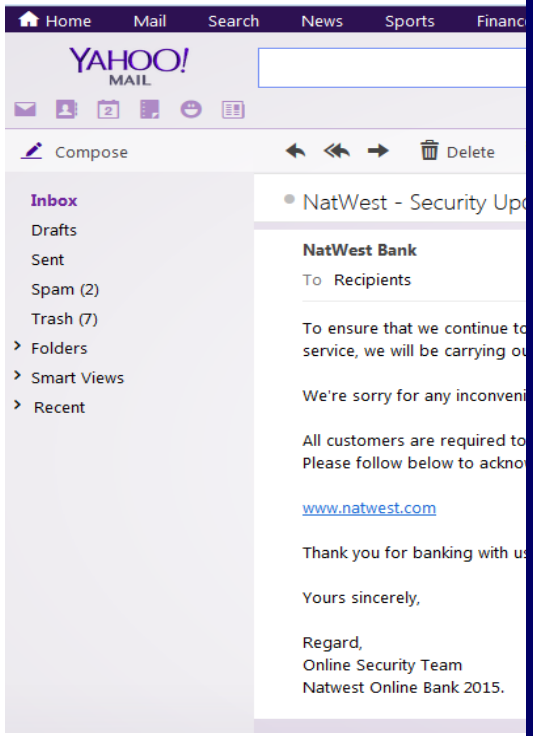
Internal Segmentation For Greater Control



Internal Segmentation For Greater Control



So What Do We Do With Dave?



Date: Thu, 15 Jan 2015 15:07:19 +0100
From: "Internal Revenue Service" <complaints@irs.gov>
To: <rpastore@cxo.com>
Subject: Complaint against your company

Dear business owner,

A criminal complaint has been filled against your company.

Your company is being accused of trying to commit tax evasion schemes.

The full text of the complaint file (PDF type) can be viewed on the IRS website, by visiting the following link :

http://www.irs.gov/complaints/view_complaint.aspx?complaint_id=931998&hash=329yt8dhui8g14

An official response from your part is required, in order to take further action.

Please review the charges brought forward in the complaint file, and contact us as soon as possible by :

Telephone Assistance for Businesses:
Toll-Free, 1-800-829-4933

Email: complaints@irs.gov

Thank you,

Internal Revenue Service

Fraud Prevention Department

Training To Raise Awareness

http://krebsonsecurity.com/2012/01/phishing-your-employees-101/

- Policy
- Procedure
- Process

Just When You Thought it Was Safe

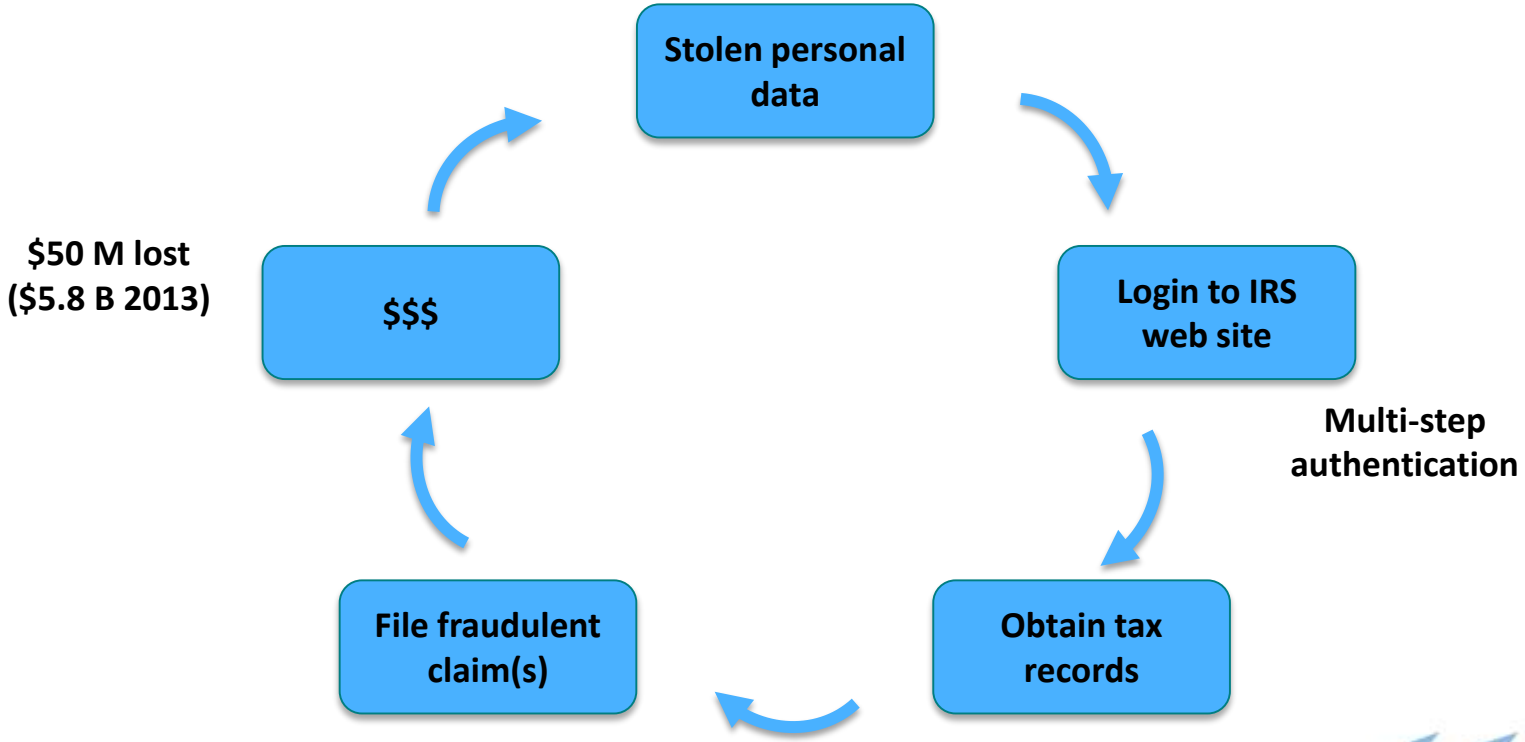
IRS

Department of the Treasury

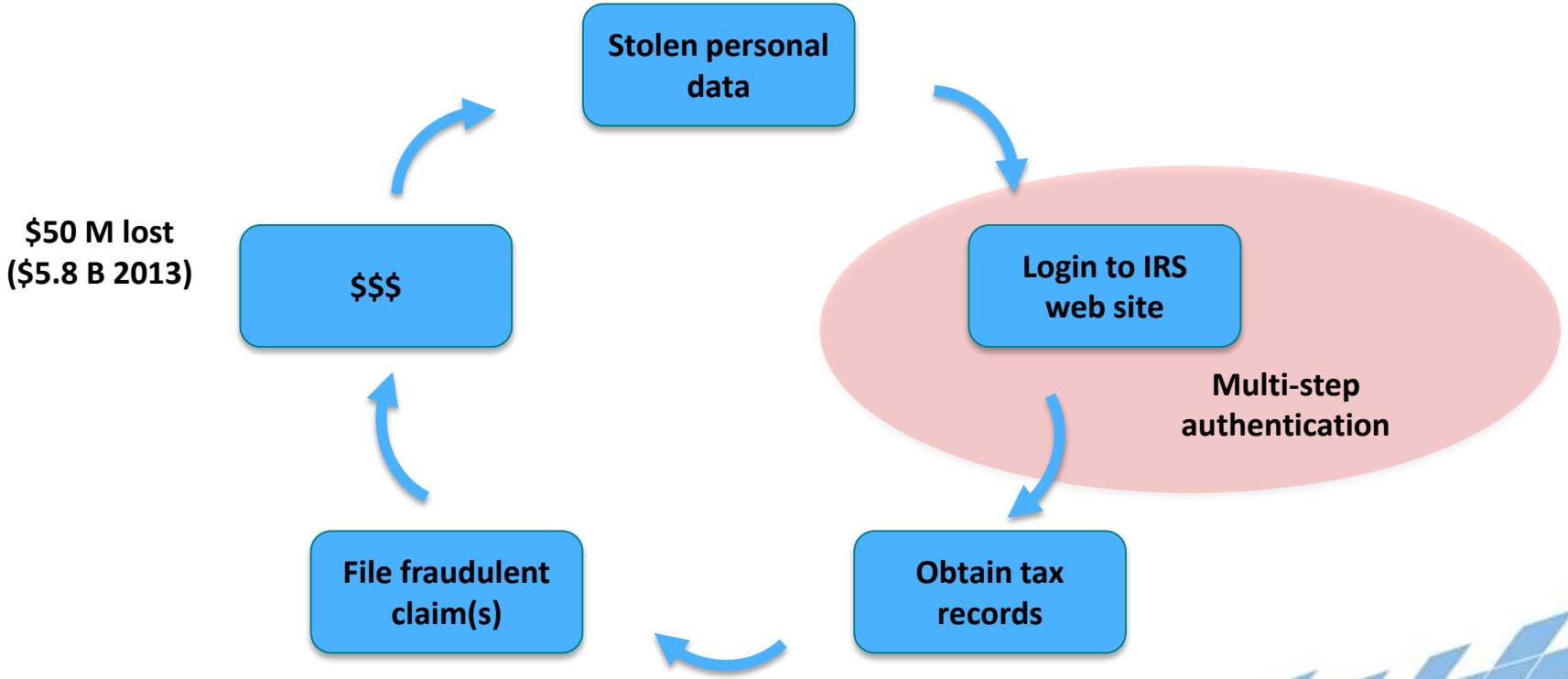
Internal Revenue Service

http://www.nytimes.com/2015/05/27/business/breach-exposes-irs-tax-returns.html?_r=0

Some But Not Enough Security



Some But Not Enough Security



200,000 attempts
(Feb – May 2015)
100,000 records accessed

Not a Good Summer for the US Government

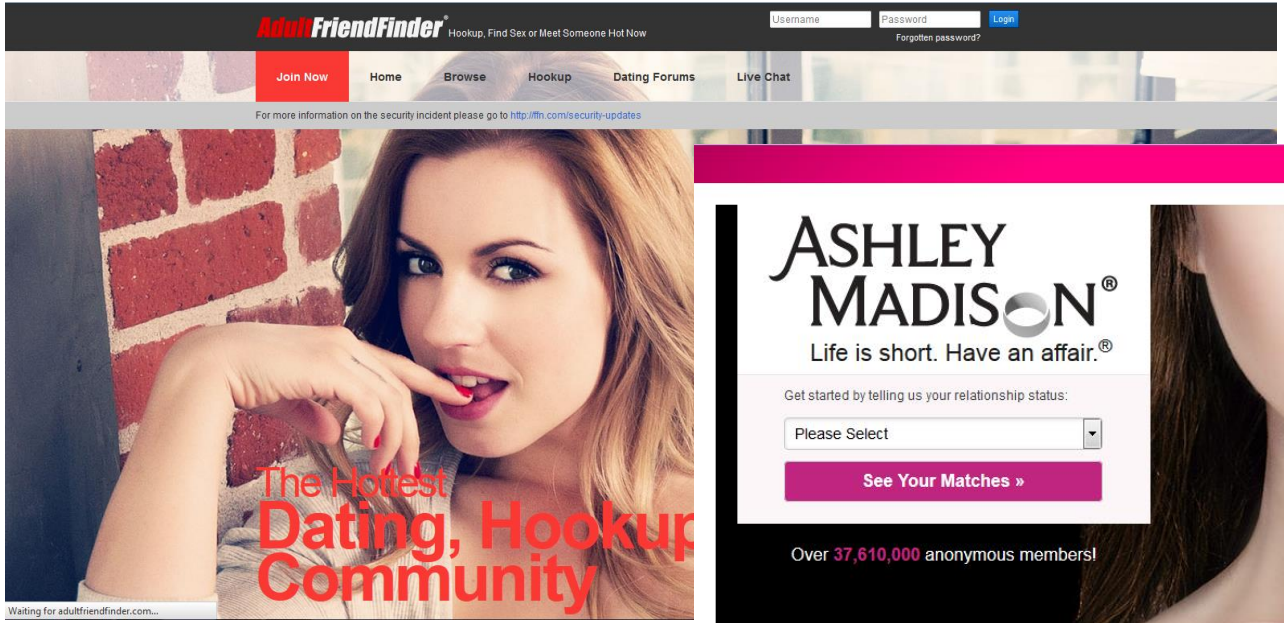
"OPM director said stolen passwords for a federal contractor were used by hackers in the two cyber attacks targeting federal employee data."

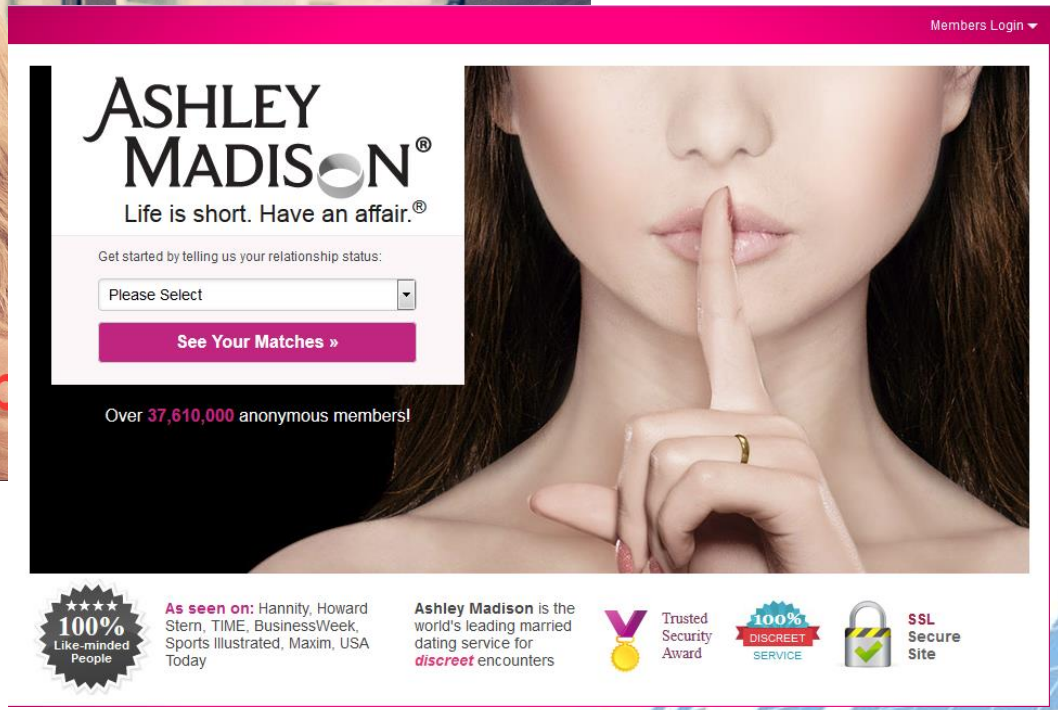
"It's really no surprise that the OPM breach was traced back to a compromised credential as this is the case in nearly 80% of the breaches we have seen, including Target and Anthem,"

Idan Tendler, head of Fortscale

http://krebsonsecurity.com/2015/06/catching-up-on-the-opm-breach/

And At the Opposite End of the Spectrum…





Looking Forward

- ✓ Technology is only part of the solution
- ✓ Consider single vendor solutions but only if elements actually work together
- ✓ Remember your network extends beyond you
  - ✓ Remote employees, third party suppliers and contractors
- ✓ Your employees are the first line of defense – Equip and use them
- ✓ Never assume
  - ✓ You're fully protected
  - ✓ The network hasn't been breached

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